

Solution Requirements

Date	2 July 2026
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** Rising Waters allows users (Disaster Management Officer) to enter weather parameters such as rainfall, cloud visibility, seasonal rainfall, etc., and predicts flood risk using a trained Machine Learning model. The system validates input data, generates flood prediction results, and displays them through a web interface.
- **Non-Functional Requirements (NFR)** Rising Waters should provide fast and accurate flood predictions, ensure secure handling of weather data, and offer an easy-to-use interface. The system should be reliable, scalable, and available whenever users need flood risk information.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in using username and password.	High
2	Authorization levels	Disaster Management Officers can enter weather data and view flood prediction results.	High
3	External interfaces	The system interacts with the trained ML model (XGBoost) and weather database.	High

4	Transactions processing	Weather data is validated, processed, and analyzed by the ML model to predict flood risk.	High
5	Reporting	Generates flood prediction results and risk reports for users.	Medium
6	Business rules, etc.	Flood prediction is based on rainfall, cloud visibility, annual rainfall, and seasonal rainfall data.	High
7	Compliance to laws or regulations	Weather and prediction data is stored securely and used only for flood prediction purposes.	High
8	<i>Other</i>	Users can instantly view flood prediction results through the web interface.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate flood predictions quickly.	Response time less than 3 seconds
2	Scalability	The system should support multiple users simultaneously.	Supports 500+ users
3	Security & Data Privacy	Weather data should be protected from unauthorized access.	Secure login and encrypted storage
4	Reliability & Availability	The system should remain available for users with minimal downtime.	99% uptime
5	Usability & Accessibility	The interface should be simple and easy to use.	User-friendly web interface
6	Other	The system should be maintainable and easy to update with new models.	Easy deployment and maintenance